

Session 4

Generative AI: retrieval and evaluation

What you build today: Project 3, the policy-grounded response service.

The job

- Answer policy questions from the current corpus, not from memory
- Cite the passage used, and mean it
- Refuse when the corpus does not cover the question
- Never show an internal document to a customer
- Prove all of that with numbers that survive a model swap

Structured output is the contract

- Everything downstream consumes a validated object, never prose
- JSON mode fails on long outputs and nested optionals
- Tool calling as a schema fails when the model declines the tool
- Schema in the prompt, validation in code, one repair: works on every vendor

Chunking is a decision

Strategy	Keeps	Loses
Fixed windows	nothing to configure	headings, metadata
Heading-aware	section, title, date, audience	little on structured docs
Sentence windows	local context	structure

Policies have headings and effective dates. Chunk on the headings, carry the metadata.

Tokenise sensitive fields first

- Email, account id, card, phone, key: replaced by a stable token
- Before the index, before the prompt, before the log
- Stable within a document so the model can still refer to "[ACCOUNT_1]"
- A test proves the values never reach a chunk

Hybrid retrieval

- Dense finds paraphrases, blurs "99.95" and "sla-2023"
- BM25 finds exact terms, misses paraphrases
- Reciprocal rank fusion merges the lists, no weight to tune
- Filter by metadata after retrieval: current only, customer audience

Reranking

- A cross-encoder reads query and chunk together
- Reorders thirty candidates, not the corpus
- Most of the quality in this session, and most of the latency
- Measure both before you keep it

Citations are enforced in code

1. Tell the model which ids it may cite
2. Check every citation against the context ids
3. Drop invented ones, count them
4. No valid citation left: refuse

Deterministic. Citation validity must be exactly 1.0 or the code is wrong.

Refusal is an outcome

Path	Reason
Retrieval too weak	<code>weak_retrieval</code> , no model call
Model says not covered	<code>model_unanswerable</code>
No valid citation survived	<code>no_valid_citation</code>

Measured against cases that must refuse, including internal-only answers.

Evaluate retrieval and generation separately

Metric	Computed by
Recall@k, MRR	deterministic
Faithfulness	the Judge, with a rubric
Citation validity	deterministic, must be 1.0
Refusal correctness	deterministic
Latency, cost	the Session 1 meter

A bad answer from good retrieval is a different bug from a bad answer from bad retrieval.

The judge is a model too

- A different, stronger model than the workhorse, fixed version
- Judges drift, prefer their own style, and can be argued with
- Spot-check a sample by hand every cohort
- Never treat a judge score as ground truth
- Calibrate it: 30 human-labelled answers, kappa and the false pass rate, `make_calibrate-judge`

The regression gate

- Write the baseline once
- Every run compares; a drop beyond tolerance fails CI
- On an engagement, the harness is what survives the vendor change

Change one variable

Around the room, one each: chunk size 200, k 2, 4 or 12, hybrid off, reranker off, min score up.

Rerun the harness. Put your row on the board. Most changes you were sure would help will not.

Acceptance criteria

1. Sensitive fields never reach a chunk; dates and percentages do
2. Superseded and internal chunks filtered by metadata
3. Citation validity exactly 1.0 on the golden set
4. Recall@8 at or above 0.85 and correct refusal on at least 15 of 17 unanswerable cases
5. A report with measured cost and p95 latency, and a passing regression gate

`make session04` reports every test passed.

Questions worth asking

- k from 8 to 2: which metric moves first, and which way?
- The judge says 0.5 on an answer you think is right. Who is right?
- The answer is only in the internal escalation matrix. What does the service say, and what does the log say?